



**RAJALAKSHMI ENGINEERING COLLEGE,
CHENNAI**

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CS23633 – Cloud Computing Laboratory**

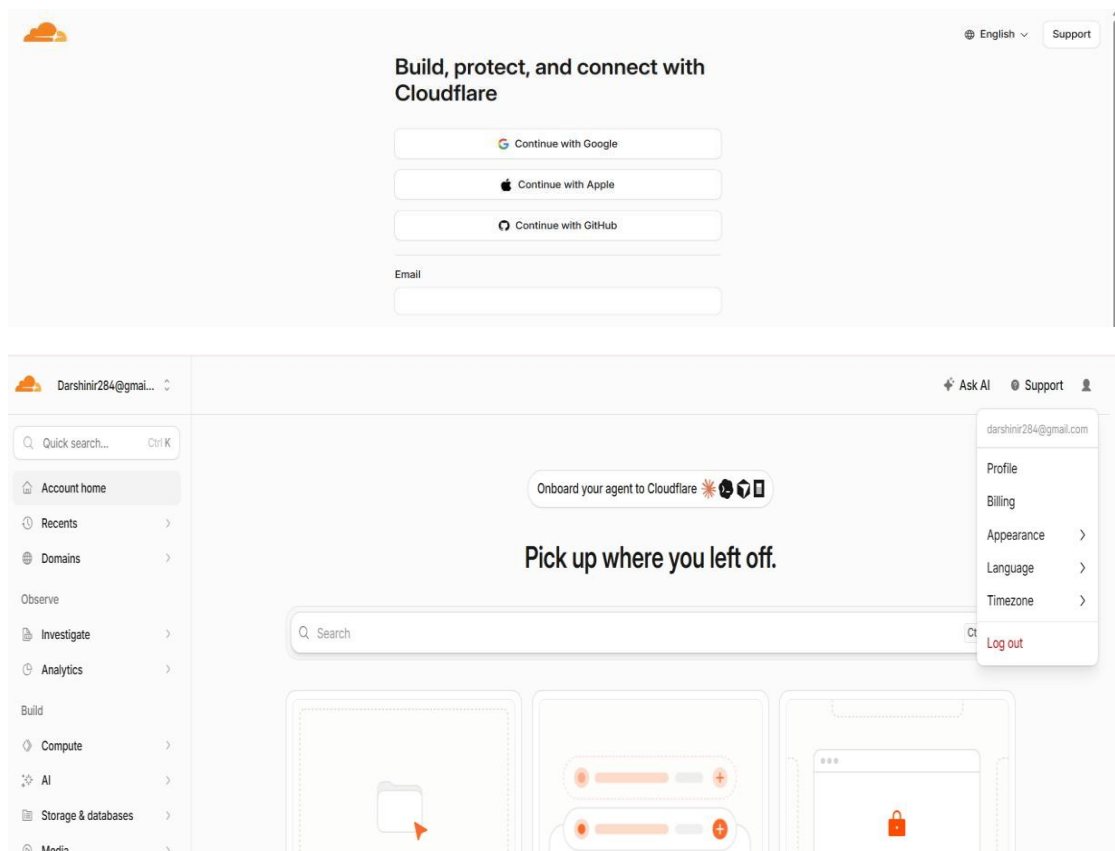
Experiment 6: CDN SERVICE SELECTION AND CDN CONFIGURATION. AIM

AIM

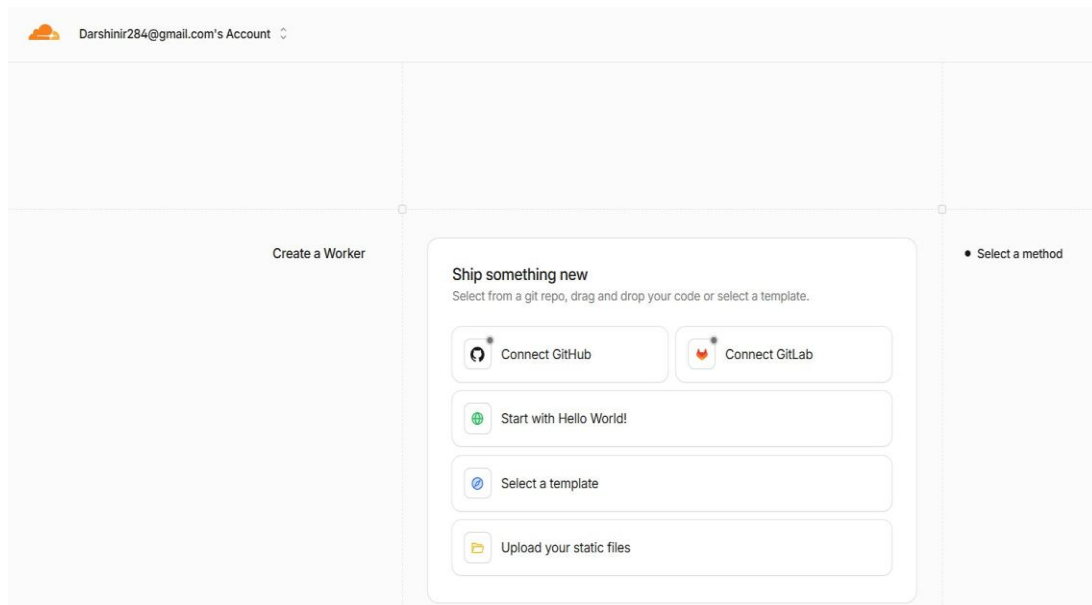
To select a Content Delivery Network (CDN) provider, deploy a simple web application using Cloudflare Workers, access the application through Cloudflare's edge network, and verify the HTTP response and Cloudflare-specific headers using browser Developer Tools.

PROCEDURE

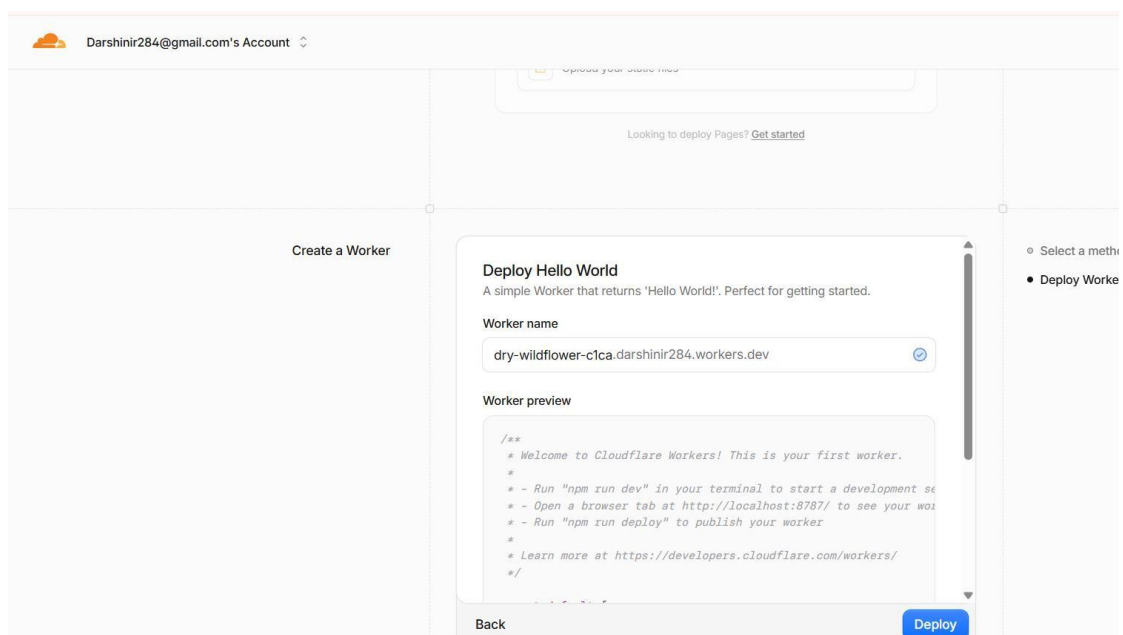
1. **Select the CDN/Edge Provider:** Cloudflare was selected as the CDN and edge computing provider because it provides CDN, security, DNS, and edge computing services.
2. **Create/Login to Cloudflare:** Open the Cloudflare website and create a new account or log in to an existing Cloudflare account. No domain or DNS configuration is required for this experiment.

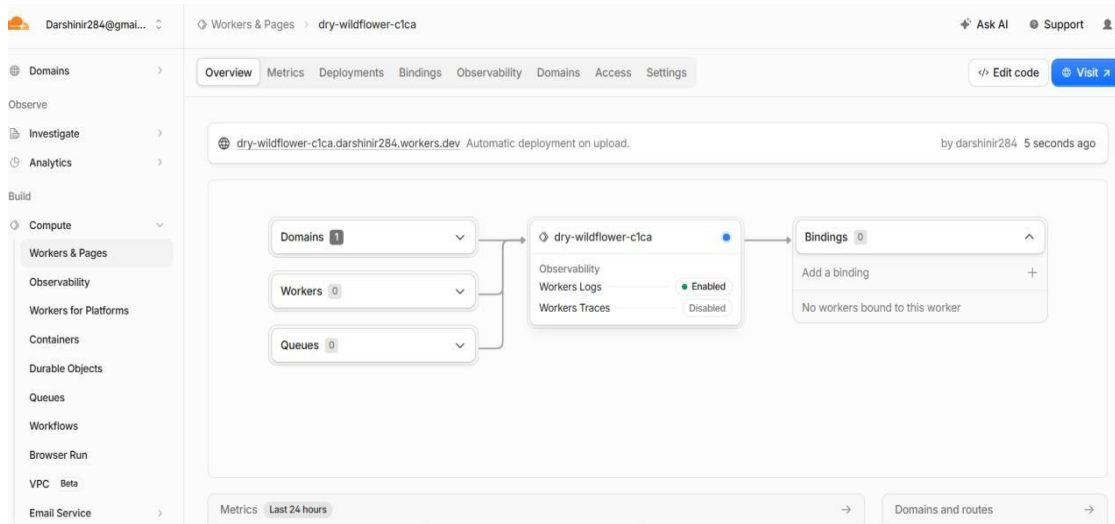


3. **Open Workers & Pages:** From the Cloudflare dashboard, navigate to **Workers & Pages** and select the option to create a new application.
4. **Create a Worker:** Select **Start with Hello World!** to create a simple Cloudflare Worker without using GitHub, GitLab, or external hosting services.

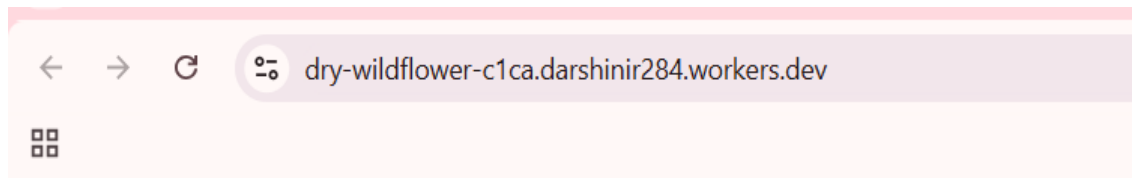


5. **Configure the Worker:** Keep the automatically generated Worker name and verify that the default Worker code returns the response Hello World!.
6. **Deploy the Worker:** Click **Deploy** and wait for Cloudflare to complete the deployment. A public workers.dev URL is generated for accessing the Worker.

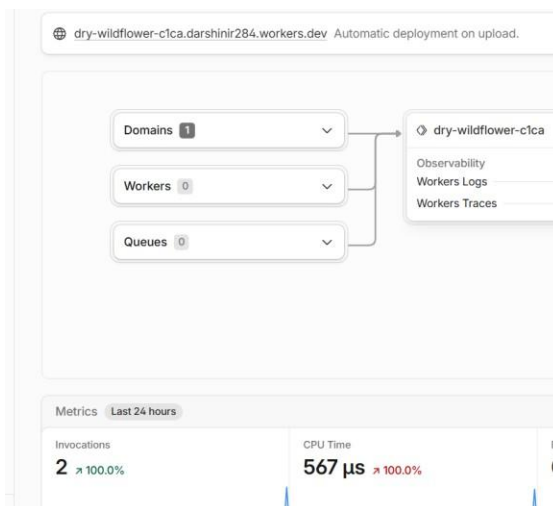




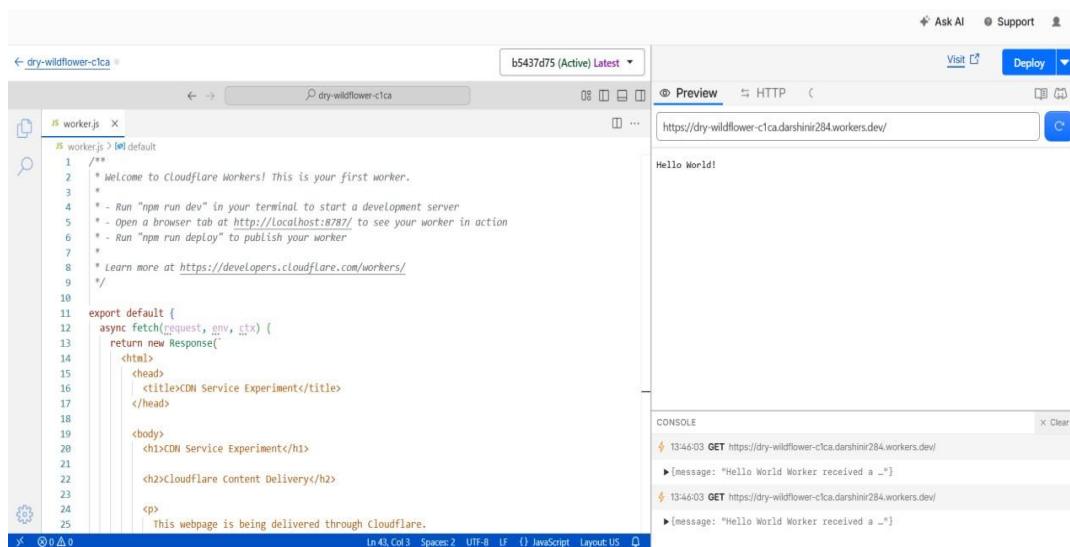
7. **Test the Deployment:** Click **Visit** or open the generated workers.dev URL in a browser. Verify that the page displays **Hello World!**, confirming that the Worker has been successfully deployed.



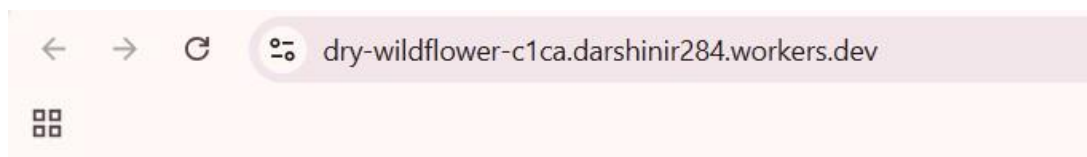
Hello World!



8. **Modify the Worker:** Open **Edit Code** and replace the default code with a simple HTML webpage containing the title **CDN Service Experiment**, information about Cloudflare content delivery, and the deployment status.



9. **Deploy the Webpage:** Save and deploy the modified Worker. Open the Worker URL again using **Visit** and verify that the newly created CDN experiment webpage is displayed correctly.



CDN Service Experiment

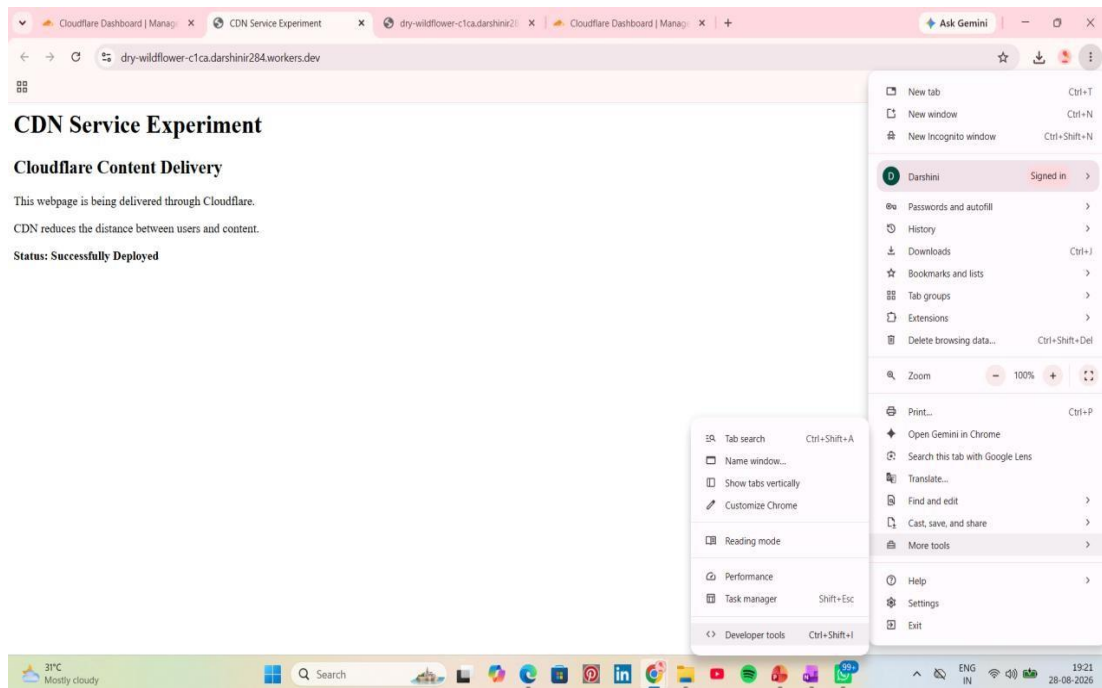
Cloudflare Content Delivery

This webpage is being delivered through Cloudflare.

CDN reduces the distance between users and content.

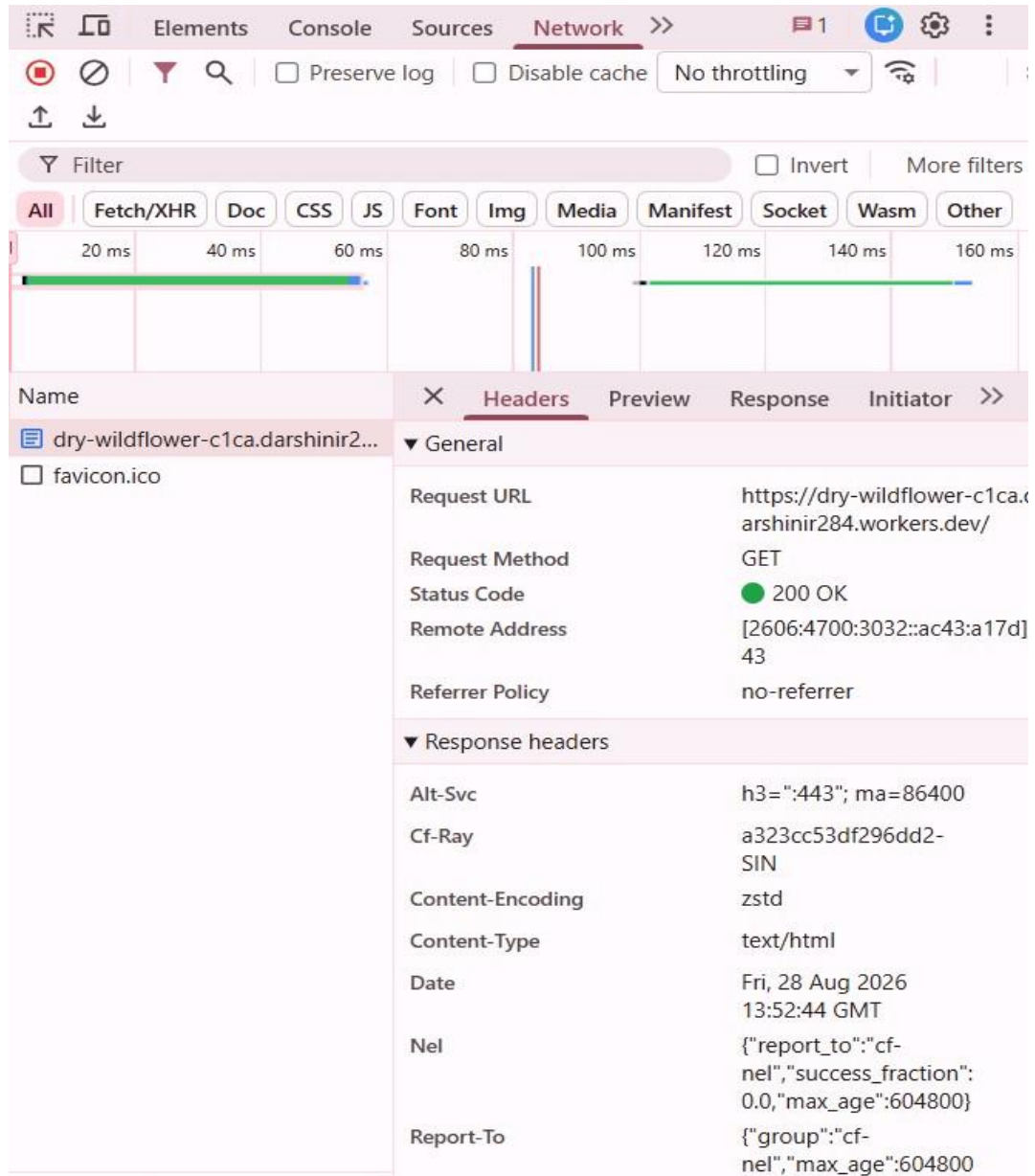
Status: Successfully Deployed

10. **Open Browser Developer Tools:** Open the Worker webpage in Google Chrome and open Developer Tools using **Ctrl + Shift + I**. Select the **Network** tab and reload the webpage using **Ctrl + R**.



11. **Inspect the HTTP Request:** Select the request corresponding to the workers.dev URL and open the **Headers** section. Verify the **Request URL**, **Request Method: GET**, and **Status Code: 200 OK**.
12. **Verify Cloudflare Response Headers:** Under **Response Headers**, verify the presence of Cloudflare-specific headers such as **Server: cloudflare** and **CF-Ray**. Also verify that the **Content-Type** is **text/html**.

13. **Record the Results:** Capture screenshots of the successful Worker deployment, the webpage, and the Network → Headers section showing 200 OK, Server: cloudflare, CF-Ray, and Content-Type: text/html. These results confirm that the application was successfully deployed and accessed through Cloudflare's edge infrastructure.



CONCLUSION

Thus, a simple web application was successfully deployed using Cloudflare Workers and accessed through a public workers.dev URL. The application returned an HTTP status code of **200 OK**, and Cloudflare-specific response headers such as **Server: cloudflare** and **CF-Ray** were observed using browser Developer Tools. Hence, the experiment successfully demonstrated the deployment and delivery of a web application through Cloudflare's edge network.

